

# AMD Ryzen 9 7950X3D

128MB of L3 Cache works its magic

**A**MD is expanding on their little experiment with 3D SRAM stacking technology and coming out with a lot more 'X3D' processors. Compared to the "vanilla" Ryzen 9 7950X, the AMD Ryzen 9 7950X3D has a lot more memory for video games to exploit. Obviously, you'd be wondering why anyone would want to get the vanilla 7950X in that case, read on to know where the vanilla processor excels.



## PERFORMANCE

### Rendering

We start off with Cinebench R23 which scores 35540 in the multi-threaded run and 2021 in the single-threaded run. In Blender, we see the Ryzen 9 7950X3D score 264, 165, and 130 in the Monster, Junkshop, and Classroom benchmarks, respectively. This is a little lesser than the AMD Ryzen 9 7950X.

### Content Creation

In the Procyon benchmarks for content creation, the 7950X3D performs a little below the 7950X and the Intel Core i9-13900K but ahead of the 7900X in the Image Retouching segment. However, in the Batch processing benchmark, it performs better than the Intel Core 13900K and the 7900X but still lags behind the 7950X. Under the video editing benchmark, the benchmark failed several times. Hopefully, this should stabilise over time.

### AI/ML

AI workloads also favour larger memory sources closer to the CPU and we saw the 7950X3D scoring higher than the 7950X by a decent margin. However, Intel has always had the upper hand in AI benchmarks. So the 7950X3D falls behind like other AMD processors. Our current AI benchmarks are purely synthetic scripts and we'll shortly be adding several real-world benchmarks.

### Compression/Decompression

Coming to compression/decompression algorithms, we use 7-Zip and WinRAR. The 7950X scores higher than the 7950X and even the i9-13900K. The same can be seen with WinRAR where the 7950X3D scores a little higher than the 7950X and the i9-13900K.

### Encryption

In our tests, Zlib encryption saw the 7950X3D was a little behind whereas in the AES and SHA3 benchmarks, the 7950X3D easily jumps ahead.

### Browser

In the Octane bench, we see the 7950X3D score a little lower than the 7950X as well as the 13900K. These benchmarks favour higher clock speeds and the 7950X3D has a lower clock, so the drop is understandable.

### Gaming

Gaming is where the 3D V-Cache really excelled with the AMD Ryzen 7 5800X3D. With a discrete graphics card connected, the scores in 3DMark Time Spy were higher than vanilla 7950X but a smidge below the i9-13900K. Whereas in the CPU Profile benchmark, we noticed that the scores were higher on the 7950X3D

when 16 Threads were running. However, when the threads dropped to 8, 4, 2, and 1, the Intel flagship had an advantage. In real-world gaming tests, Riftbreaker is where we saw the highest delta between the 7950X and the 7950X3D in favour of the 3D variant. In games such as Far Cry 6, the 7950X3D is neck and neck with the 13900K. In F1 2022, the 13900K is a little ahead of the 7950X3D. In Witcher 3: The Wild Hunt, the 13900K retains its lead. Shadow of the Tomb Raider reverses that with the AMD Ryzen 9 7950X3D inching ahead. So we've seen quite a lot of movement in the video game performance benchmarks with the AMD Ryzen 9 7950X3D taking the crown away from Intel in a lot of titles while not having

much success in some titles. Overall, the 7950X3D is an equal if not a better gaming processor. In terms of power, the 7950X3D consumes 125 W on average and in high-demand scenarios peak levels hit 145 watts.

## VERDICT

Now should get the 7950X or the 7950X3D? Well, if your usage is going to be gaming-heavy, then go for the 7950X3D. The 7950X still has a place thanks to its higher thermal headroom. Considering that there are trade-offs between the 7950X and the 7950X3D (incl. the higher price in India), you can't say that one processor is better than the other across all workloads and therefore, can't command a higher price. So at the price of INR 70,799, you're better off with the vanilla AMD Ryzen 9 7950X.

—Mithun Mohandas

### SPECIFICATIONS

Socket: AM5 | Core Process: 5 nm | Core Architecture: Zen 4 | Die Size: 71 mm<sup>2</sup> | Cores: 16 - Big Cores 16 - Small Cores 0 Threads 32 | Integrated Graphics: RDNA2 Base Multiplier: 42x | Memory Type Support: DDR5 | Max Memory Capacity: 128 GB | Rated Memory Clock Speed: 5200 MT/s | Memory Channel Support: Dual-Channel | PCIe Gen: 5 | PCIe Lanes from CPU: 24 | TDP: 120 W

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